Input, output and storage devices

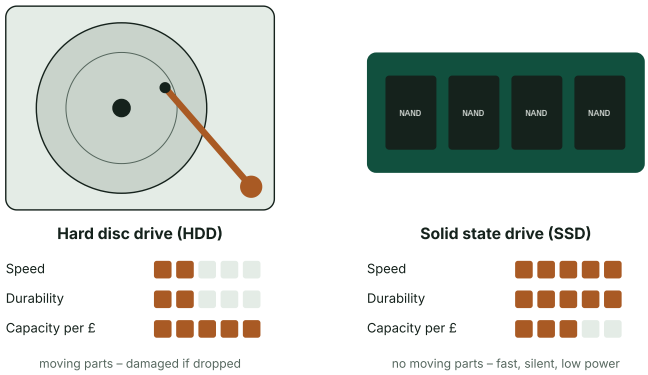
L15 FACT FILE

Specification: students should be able to

- Describe the characteristics, typical uses, and advantages and disadvantages of: microphone, mouse, graphics digitiser, touch screens, speakers, printers (laser and 3D), hard disc drive (HDD), high definition (HD) storage media and solid state drive (SSD)
- Describe other solid-state storage: USB flash drives, memory cards and smart cards
- Compare storage devices using capacity, access speed, portability, durability and cost
- Recommend and justify suitable devices for a given scenario

CCEA fact file: Fact File 11: Computer Hardware

Key words: input device, output device, storage device, microphone, voice recognition, graphics digitiser, stylus, touch screen, laser printer, toner, 3D printer, CAD, HDD, SSD, flash memory, Blu-ray, capacity, durability



Hard disc drive with spinning platter compared with a solid state drive made of memory chips

Devices and why they are chosen

Hardware devices are grouped by what they do. Input devices get data into a computer: a microphone captures sound, a mouse controls the pointer, a graphics digitiser captures freehand drawing with a stylus, and a touch screen lets users select options by touching the display. Output devices present information: speakers produce sound, a laser printer produces high-quality pages, and a 3D printer builds solid objects. Storage devices keep data permanently, even when the power is off: hard disc drives store data magnetically, solid state drives use flash memory, and high-definition optical discs such as Blu-ray use a laser. A touch screen is unusual because it is both an input and an output device. No device is best in every situation. A device is chosen by matching its features to the task, the users and the environment. A microphone is ideal for a gamer chatting to teammates, but voice input would be a poor choice in a noisy school canteen. Exam questions nearly always give a scenario, so every advantage you give should be linked to it.

Input devices

A microphone turns sound into an electrical signal that is converted into digital data. It is used for voice recognition, such as dictating text or giving commands to a smart speaker, as well as for video calls, recording music and online gaming headsets. Speaking is faster than typing and is hands-free, but background noise reduces accuracy and sound files need much more storage than text. A mouse uses sensors to detect movement and moves the on-screen pointer. It is easy to learn, but needs a flat surface, and experienced users often find keyboard shortcuts quicker. A graphics digitiser is a flat pad used with a stylus, much like a pencil on paper. It gives more natural and more accurate drawing than a mouse, which suits artists and designers, and it can also capture signatures. However, it is more expensive than a mouse and poor for selecting menus. A touch screen has a touch-sensitive glass surface. A touch changes the electrical signal, and the position is worked out as x, y co-ordinates. Touch screens need little skill and are quick, but they get dirty and can only show a limited number of options.

Output devices

Speakers turn electrical signals from the sound card into sound. They are used for music, video, games and alert sounds, and they help visually impaired users by reading text aloud. Their main drawback is that they can disturb others nearby, so headphones may be better in shared spaces. A laser printer receives the page as data. A laser beam draws the page onto a drum as a pattern of static charge, powdered toner sticks to the charged areas, and the toner is transferred to paper and fixed with heat. Laser printers are fast for large volumes and give sharp text and graphics, which is why schools and offices use them. However, colour models are expensive to buy and run, and because they are non-impact printers they cannot print multipart carbon-copy forms. A 3D printer builds a solid object one layer at a time from a CAD design, using materials such as plastic, metal or ceramics. It is ideal for prototypes and custom items such as prosthetic hands, but it is slow for large objects and not yet economical for mass production.

Storage devices

A hard disc drive (HDD) contains rigid magnetic discs, called platters, sealed in a case to keep dust out. Each surface is split into circular tracks and the tracks into sectors, and read/write heads move across the spinning discs to reach data. HDDs offer very large capacities at the lowest cost per gigabyte, but their moving parts can fail or be damaged if dropped, and access is slower than flash storage. A solid state drive (SSD) uses flash memory with no moving parts. It is non-volatile, much faster, quiet, light, uses less power and survives knocks, but costs more per gigabyte. Other flash storage includes USB drives, memory cards for phones and cameras, and the smart-card chips in bank cards and door entry cards. High-definition storage media, such as Blu-ray, are optical discs. A blue-violet laser has a shorter wavelength than the red laser in a DVD drive, so the pits are smaller and packed more tightly, and a dual-layer disc holds 50 GB. They suit HD films and games, but scratch easily.

T Key terms from the mark schemes

Input device
Hardware used to enter data into a computer, such as a microphone, mouse or graphics digitiser.

Output device
Hardware used to present information from a computer, such as speakers or a printer.

Graphics digitiser
An input device made of a flat pad and a stylus, used to capture freehand drawings or signatures.
`Credited: Graphics tablet`

Touch screen
A display with a touch-sensitive surface that works as both an input and an output device.
`Credited: Touchscreen`

Laser printer
An output device that uses a laser to form the page on a charged drum, which attracts toner that is fused onto paper; fast and high quality.

3D printer
An output device that builds a solid object layer by layer from a computer-aided design (CAD) model.
`Credited: Additive manufacturing`

Hard disc drive (HDD)
Magnetic storage using spinning platters and moving read/write heads; large capacity at low cost per gigabyte.
`Credited: HDD`

Solid state drive (SSD)
Non-volatile flash storage with no moving parts: faster, more durable and lower power than an HDD, but more expensive per gigabyte.
`Credited: SSD`

Optical storage
Storage read and written by a laser, such as Blu-ray, the HD storage media named in the specification.
`Credited: HD storage media`

Flash memory
Non-volatile electronic memory with no moving parts, used in SSDs, USB drives and memory cards.
`Credited: Solid-state storage`

Smart card
A plastic card with an embedded chip that stores a small amount of data, such as a chip-and-PIN bank card.

Exam focus

Devices are tested often and usually in a context. Input devices and their uses appeared in 2023 Q5(e), 2024 Q9(a) and 2025 Q7(c). Laser printer and SSD advantages were tested in 2021 Q7, and an SSD advantage in 2019 Q5(a). 3D printing and graphics digitisers came up in 2022 Q5, and optical storage and capacity terms in 2022 Q2 and the 2022 Q1 multiple choice. Marks are given for a valid point that is clearly explained and linked to the scenario. 'An SSD is faster' may get one mark; 'an SSD has no moving parts, so the inspector's laptop is less likely to be damaged if dropped' gets more. Make your points comparative, such as 'lower cost per gigabyte than an SSD'. Do not give the same idea twice in different words, and avoid vague answers such as 'it is better' or 'it is easier'. Learn the key facts: 3D printers work layer by layer, laser printers use toner, HDDs are magnetic and SSDs have no moving parts.

Unit 1 2025 Q13	[10]	Term matching: hardware, memory, networking, malware
Unit 1 2025 Q7	[8]	Fetch–execute cycle; input devices and uses
Unit 1 2024 Q9	[7]	Input devices, health risks, gaming and internet risks
Unit 1 2023 Q5	[10]	CPU, system software resources, real-time vs multi-user, utilities, I/O devices
Unit 1 2022 Q5	[6]	3D printing, utilities and security, graphics digitiser

? Check yourself

1. State whether each of the following is an input device, an output device or both: microphone, speakers, touch screen. [3]
2. Describe one advantage and one disadvantage of using a touch screen to take orders in a fast-food restaurant. [2]
3. Describe what a graphics digitiser is and give a typical use. [2]

Answers: 1. Microphone – input (1) / speakers – output (1) / touch screen – both (1). 2. Advantage, one from: customers or staff need little training / quick to select items / no separate keyboard or mouse (1). Disadvantage, one from: screen gets dirty or greasy, a hygiene issue (1) / limited number of options on screen at once / can be damaged with heavy use. 3. A flat pad/tablet used with a stylus to capture freehand drawing (1). Use: graphic design / artwork / animation / capturing signatures (1).